Tasks and Deadlines

Time limit: 1.00 s **Memory limit:** 512 MB

You have to process n tasks. Each task has a duration and a deadline, and you will process the tasks in some order one after another. Your reward for a task is $d - f$ where d is its deadline and f is your finishing time. (The starting time is 0, and you have to process all tasks even if a task would yield negative reward.)

What is your maximum reward if you act optimally?

Input

The first input line has an integer n: the number of tasks.

After this, there are n lines that describe the tasks. Each line has two integers a and d: the duration and deadline of the task.

Output

Print one integer: the maximum reward.

Constraints

- $1 \le n \le 2 \cdot 10^5$
- $1 \le a, d \le 10^6$

Example

Input	Output
3 6 10 8 15 5 12	2